

SE 216 – SOFTWARE PROJECT MANAGEMENT

SOFTWARE MEASUREMENTS DOCUMENT

PROJECT NAME: BioPulse : Autonomous Green Ops Platform

GROUP NUMBER and MEMBERS: GROUP 2 / Umutcan Yavuz, Kaan Pamuk, Erdi Sinan, Ahmet Yiğit Sertel, Ata Çelik, Onur Özbek, Enes Görkem Hasgör

Questions to identify measurements:

Note: Please be advised that the ensuing inquiries have been meticulously correlated with their respective fundamental measurements in a sequential and systematic order.

- 1. What is the data processing latency from sensor input to system response?**
- 2. What is the data loss rate during transmission or storage?**
- 3. Did the project adhere to the planned schedule and sprint deadlines?**
- 4. How effective is the alert and autonomous decision-making mechanism?**
- 5. How well does the system meet quality requirements (e.g., performance, accuracy, reliability)?**
- 6. How much effort was dedicated specifically to testing and validation activities?**
- 7. How efficiently is sensor data transmitted and stored in the database?**
- 8. How fast are the batteries depleting?**

Identified measurements:

The following **8** fundamental metrics have been constructed in order to fully assess the BioPulse platform in light of the specified questions:

- 1. End-to-End System Latency:** Determines how long it takes for an autonomous system to activate after receiving physical sensor input.
- 2. Telemetry Packet Delivery Rate (PDR):** Measures data loss during Wi-Fi/SIM routing and assesses the effectiveness of data transmission.
- 3. Variance in the Sprint Schedule (SV):** Evaluates compliance with the four primary sprints of the Agile calendar.
- 4. Autonomous Logic Precision:** Uses sensor data and weather forecasts to calculate the true-positive versus false-positive rate of irrigation decisions.
- 5. System Uptime & Availability:** Monitors the database servers' and Basic Backend API's operational dependability.
- 6. Test Automation Coverage:** Calculates the proportion of codebase that is run during automated testing to quantify the work put into validation.
- 7. Database Query Execution Time:** Tracks MongoDB's efficiency when retrieving and storing large amounts of telemetry data.

SE 216 – SOFTWARE PROJECT MANAGEMENT

SOFTWARE MEASUREMENTS DOCUMENT

8. Hardware Power Drain Rate: Evaluates the energy efficiency of distant nodes that are operating in "Low power consumption mode."

Measurement storage and collection:

1. End-to-End System Latency

- **What:** The time difference between the sensor input and resulting system action
- **When:** Every time an event occurs.
- **Where:** MongoDB `latency_metrics` collection.
- **How/Who:** The backend API automatically records timestamps. Developers (Umutcan, Kaan, Ahmet, Ata, Onur) monitor these logs to ensure the system meets performance requirements.

2. Telemetry Packet Delivery Rate (PDR)

- **What:** Missing sequence counts and the delivered/sent ratio for Wi-Fi and SIM channels.
- **When:** Daily aggregation.
- **Where:** MongoDB `transmission_log` collection.
- **How/Who:** The Raspberry Pi gateway automatically computes the ratio. The Development Team reviews the logs to optimize connectivity in the field.

3. Sprint Schedule Variance

- **What:** The planned completion date versus the actual completion date of sprint tasks.
- **When:** At the end of every sprint.
- **Where:** Shared project tracking spreadsheet.
- **How/Who:** Scrum Master (Enes Görkem Hasgör) calculates the variance by comparing planned and actual progress to identify potential bottlenecks.

4. Autonomous Logic Precision

- **What:** Counts of correct decisions, false positives, and incorrect decisions regarding autonomous actions.
- **When:** During system testing scenarios.
- **Where:** Testing spreadsheet and test report.
- **How/Who:** The Developers execute the tests, while Product Owner (Erdi Sinan) validates the results to ensure the logic aligns with user needs.

5. System Uptime and Availability

- **What:** Total system running time, downtime, and availability percentage.
- **When:** Continuously monitored.
- **Where:** Server/API monitoring logs.
- **How/Who:** Automated monitoring tools record uptime. Scrum Master (Enes Görkem Hasgör) reviews these reports during retrospective meetings to discuss infrastructure stability.

SE 216 – SOFTWARE PROJECT MANAGEMENT

SOFTWARE MEASUREMENTS DOCUMENT

6. Test Automation Coverage

- **What:** The percentage of the BioPulse codebase covered by automated tests.
- **When:** After major code updates and before every sprint review.
- **Where:** Automated test report folder.
- **How/Who:** The Development Team (Umutcan, Kaan, Ahmet, Ata, Onur) is responsible for writing and running the test suites to maintain high code quality.

7. Database Query Execution Time

- **What:** Execution time for critical database queries.
- **When:** Every query execution (via profiler).
- **Where:** MongoDB `db_perf_metrics` collection.
- **How/Who:** The MongoDB profiler logs times automatically. Developers analyze these metrics to optimize database performance and indexing.

8. Hardware Power Drain Rate

- **What:** Battery percentage, voltage levels, and milliwatt-hour usage.
- **When:** During long-running field tests and stress tests.
- **Where:** Hardware test log and shared measurement spreadsheet.
- **How/Who:** The Development Team manually records and analyzes power consumption to ensure the IoT nodes meet sustainability goals.

Measurement Type	Description	Example Measurements
End-to-End System Latency <i>(Performance Metric)</i>	Measures how quickly the system reacts after receiving sensor data.	Average latency < 500ms; Maximum peak latency < 2.0 seconds.
Telemetry Packet Delivery Rate <i>(Reliability Metric)</i>	Measures how reliably sensor data is transmitted without packet loss.	99.9% successful delivery rate; < 0.1% packet loss rate.
Sprint Schedule Variance <i>(Project Management Metric)</i>	Measures how closely the project follows the planned sprint timeline.	SV = 0 (Exactly on schedule); SV = -2 days (Delayed by 2 days).
Autonomous Logic Precision <i>(Quality/Accuracy Metric)</i>	Measures how accurately the system makes irrigation or fertilization decisions.	Target precision ≥ 95%; false positive rate < 5%.
Test Automation Coverage <i>(Quality Assurance Metric)</i>	Measures how much of the code is tested automatically during validation.	At least 90% coverage of critical irrigation logic and 80–85% overall code coverage.

SE 216 – SOFTWARE PROJECT MANAGEMENT
SOFTWARE MEASUREMENTS DOCUMENT

System Uptime & Availability <i>(Performance Metric)</i>	Measures how consistently the API and system services remain accessible and operational.	99.99% monthly uptime with less than five minutes of unplanned downtime.
Database Query Execution Time <i>(Efficiency Metric)</i>	Measures how efficiently the database performs read and write operators.	Average write time < 50 ms; average database read time < 100 ms.